A Further Discussion

Broader impact. This paper presents work that aims to advance the field of trustworthy machine learning and large language models. We do not find any negative societal consequences of our work. This paper does not raise any ethical concerns. This study does not involve human subjects, practices, data set releases, potentially harmful insights, methodologies, applications, potential conflicts of interest and sponsorship, discrimination/bias/fairness concerns, privacy and security issues, legal compliance, or research integrity issues.

Limitations. The proposed CD-CoT method is currently dependent on human-annotated, clean rationales. Future enhancements could include developing a self-supervised variant that does not rely on such examples. Exploring strategies like using contrasting noisy examples or incorporating an external knowledge base, possibly through a retrieval-augmented denoising framework, may offer significant advances in automation and robustness of reasoning.

Extensions. CoT and its variants have predominantly focused on deductive reasoning, leaving inductive reasoning largely unexplored. Investigating the ability of LLMs to extract rules from noisy examples is a compelling area. Additionally, theoretical analysis of noisy ICL can offer deeper insights into the Noisy-R problem. Expanding the NoRa dataset to include multi-modal scenarios, particularly visual data, is also crucial for a more comprehensive understanding of the robustness of foundation models. Further extensions include knowledge-enhanced denoising [100, 46, 47, 113], generalization to out-of-distribution noisy scenarios [7, 8], and training to fundamentally improve the robustness of language models [75, 74].

B Related Work

In this section, we provide a detailed literature review as an extension of the preliminaries (Sec. 2), including in-context learning (Appendix B.1), self-correction methods (Appendix B.2), self-consistency methods (Appendix B.3), and external supervision (Appendix B.4). We further discuss the relation between our work and literature in Appendix B.5. We also provide Fig. 6 to better understand different reasoning settings.

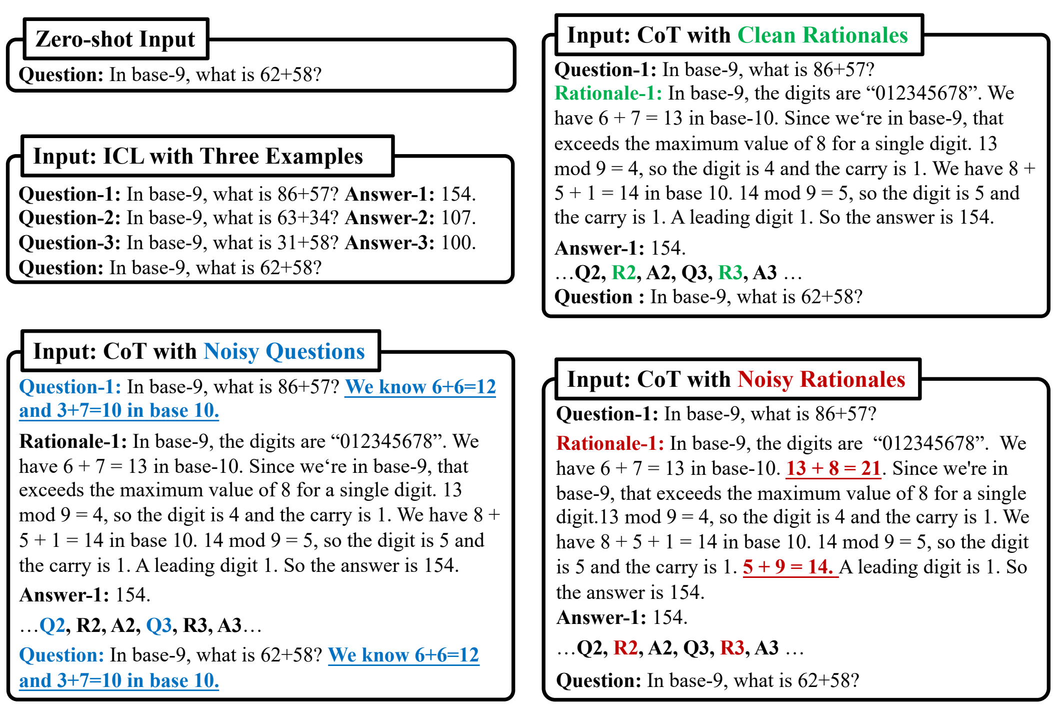

Figure 6: Illustrations of different reasoning settings.

B.1 In-context Learning

With the scaling of model size and corpus size [61, 4, 11, 105], large language models (LLMs) have demonstrated remarkable performance across a variety of tasks through in-context learning (ICL), that is, learning from a few input-output examples in the context. As a training-free framework, ICL can save on expensive training costs and be easily adapted to diverse new tasks [88, 1].

However, standard ICL faces challenges when addressing *intricate* reasoning problems. Wei et al. [85] proposes chain-of-thought prompting (CoT), a methodology that explicitly guides LLMs in generating sequential reasoning steps to enhance their performance on intricate reasoning tasks. In particular, CoT replaces the input-output exemplars in ICL with triplets in the form of <*input, rationale, output*>, thereby enabling the model to learn explicit reasoning processes.

In the literature, various versions of chain-of-thought prompting have been proposed. To alleviate the human effort required for constructing triplet exemplars, Kojima et al. [39] proposes zero-shot CoT, a method that explicitly encourages models to generate a chain of reasoning first and then derive the final answer by employing natural language prompts like "Let's think step by step". Wang et al. [83] shows that self-consistency sampling can improve reasoning accuracy by sampling diverse reasoning paths and then taking the majority vote. Zelikman et al. [97] proposes STaR, which leverages a small amount of human-annotated rationale data, along with a large dataset of question-label pairs without rationales. This approach iteratively generates additional rationales and enhances the model's ability to generate reasoning steps by fine-tuning.

Least-to-most prompting (LtM) [109] enhances the reasoning capability of models by decomposing a given problem into sub-problems. In particular, LtM initially breaks down the task into a series of consecutive sub-problems and subsequently answers them one by one. During the process of responding, the answer to the preceding sub-problem is incorporated into the prompt for the succeeding one. Tree of Thoughts (ToT) [92] extends LtM by exploring multiple reasoning possibilities at each step. Specifically, ToT first decomposes a given problem into several reasoning steps and generates multiple answers for each step, ultimately constructing a tree structure. Subsequently, ToT employs BFS or DFS to traverse the tree, yielding the final rationale and answer.

Vulnerabilities of ICL. Despite being promising, some works point out the brittleness and over-sensitivity of ICL. Liu et al. [50], Perez et al. [60], Zhang et al. [99] demonstrate that ICL performance depends heavily on the choice of exemplars. Meanwhile, Zhao et al. [106], Lu et al. [51] observe that the arrangement order of in-context examples is also crucial to the ICL performance, potentially shifting results from near state-of-the-art to a random guessing. Moreover, Ye and Durrett [94], Gan and Mori [19], Zheng and Saparov [107], Zhang et al. [102] reveal LLMs' deficiencies when handling subtle perturbations within the prompts, even when such perturbations do not alter any semantic meaning. Yu et al. [95] explores the robustness of retrieval-augmented in-context learning (ICL) against demonstration attacks and test sample attacks. It focuses on perturbing the example questions (i.e., noisy questions) or labels, while our work focuses on the rationales of the examples (i.e., noisy rationales). In addition, previous work on safety [44, 112] and data noise [24, 49, 22, 111] might also inspire the robust problems in ICL. Overall, the enhancement of reasoning performance brought about by ICL is inherently unstable and susceptible to example selection, example ordering, and prompt perturbations. These observations underscore the importance the robustness of other aspects.

The aforementioned efforts primarily revolve around the idealized ICL, which utilizes high-quality prompts free from any noise or interference. Conversely, a parallel line of research has emerged, exploring the impact of noisy prompts on the performance of LLMs. Min et al. [54] examines the impact of in-context examples on ICL. This work observes that incorporating out-of-distribution input texts significantly diminishes the performance of standard question answering. Wei et al. [86] devises two different set-ups of ICL: ICL with flipped labels and ICL with semantically unrelated labels. Their investigation reveals that LLMs possess the capability to override semantic priors when confronted with in-context exemplars that contradict these priors. This phenomenon also suggests that larger models may be more susceptible to the influence of the noise present in examples. Shi et al. [68] examines the impact of irrelevant context on LLMs, and the results suggest that the inclusion of irrelevant information can significantly impair the performance of the models. These studies further illuminate the fragility and instability inherent in the reasoning capabilities of LLMs.

However, the previous works mainly consider the noisy questions/answers in standard ICL. In contrast, we move to the under-explored noisy rationales problem in the context of CoT, as illustrated in Fig. 6.

Numerous strategies have been proposed to address the vulnerabilities of LLM reasoning during in-context learning. These approaches can be categorized into *self-correction* and *self-consistency*, which are introduced as follows.

B.2 Self-correction

Self-correction emerges as a promising direction to enhance LLM reasoning, where LLMs attempt to correct their initial responses based on feedback. One popular line of research involves utilizing manual labor or external systems to evaluate and refine models. However, this can be costly due to the manual labor involved. Another line of research leverages the LLM's inherent capabilities to correct its initial responses without the crutch of external feedback. This methodology is a promising way to make LLM-based solutions practical and deployable [57].

Self-correction with internal feedback. In this line of research, the LLM is required to correct response trajectories based solely on its inherent capabilities. Huang et al. [28] first demonstrates the self-improvement potential of LLMs by utilizing a pre-trained LLM to generate rationale-augmented answers for unlabeled questions using CoT and majority voting and then fine-tuning the LLM using those self-generated labels, eventually improving the general reasoning ability.

When addressing problems, people typically engage in trial and error, coupled with reflective thinking, to discern the correct solutions. Inspired by this, Madaan et al. [52] proposes Self-refine, a simple and direct approach to improving LLM's output. In this approach, an LLM is used to create an initial output. Then, the model provides feedback on its own output in multiple dimensions. Based on this feedback, the model refines its initial output and repeats this process until it reaches a specified limit or the LLM determines that no further adjustments are necessary.

Encouraged by the augmented efficacy achieved through self-feedback mechanisms, Ye et al. [93] releases SelFee, a new instruction-following language model that generates self-feedback on its response and self-revises based on the feedback. The development of SelFee involves the fine-tuning of LLaMA by utilizing training instances generated by ChatGPT.

In addition, Gero et al. [21] introduces Self-verification, suggesting that by asking LLMs to provide provenance for their own outputs and conducting checks, it is possible to alleviate LLMs' issues regarding accuracy and interpretability in crucial domains such as healthcare. On the other hand, Xi et al. [89] focuses on the simplicity and comprehensibility of the given questions, proposing Self-polish (SP). This method instructs the LLM to iteratively refine the test question by removing irrelevant information and rearranging the logical structure, thereby improving the reasoning performance.

While the self-correction methodologies based on internal feedback appear promising, [29] categorizes such self-correction methods as *intrinsic self-correction* (ISC) and demonstrates that the model's performance drops on all benchmarks after using ISC. This work points out that LLMs struggle to self-correct their responses without external feedback, and the corrected responses often exhibit inferior quality compared to their initial counterparts. Saparov and He [65] reveals that while models are able to produce valid reasoning steps with high probability when dealing with proof problems, they struggle with proof planning. In other words, when models occasionally generate incorrect proof steps, they are not able to return to the correct path.

Building upon this observation, Tyen et al. [81] further decomposes the self-correction process into two core components: mistake finding and output correction. This work demonstrates that current state-of-the-art LLMs cannot find mistakes reliably, even in the most simple and unambiguous cases, and suggests this is a main contributing factor to LLMs' inability to self-correct reasoning errors.

Self-correction with external feedback. External feedback offers a valuable external perspective, proving particularly advantageous in pinpointing errors that the large language model may not inherently recognize [57]. The sources of external feedback can be categorized as 1) human feedback [67, 69, 36, 81], 2) external tools [34, 96, 43], and 3) other models [59, 3].

Scheurer et al. [67] proposes Imitation Learning from Language Feedback (ILF), an approach leveraging informative human feedback that involves conditioning the model on input, initial output, and feedback; selecting the most feedback-incorporated refinement; and fine-tuning the model to maximize the chosen refinement's likelihood given the input.

Similarly, Shinn et al. [69] introduces Reflexion, which fortifies language agents by relying on linguistic feedback generated by themselves rather than weight updates, resulting in noteworthy enhancements compared to a baseline agent across a spectrum of tasks. Kim et al. [36] demonstrates